lower solubility, providing a seeding effect to Li_2CO_3 nucleation. Li_2CO_3 nucleated and grew on the generated $MgCO_3{\cdot}xH_2O$ seeds as an octahedron-like morphology, which is highly supposed to have resulted from the $MgCO_3{\cdot}xH_2O$ seeds[21]. Morphology control is a critical factor in enhancing flowability and compressibility during wet-chemical powder production. As a result, numerous studies in lithium production have focused on achieving well-defined particle morphologies by reducing supersaturation[22–24] and adjusting crystal growth conditions[25]. Our observation suggests that Mg impurity is effective in controlling morphology as well, thus preventing lithium powders from clogging in the reaction batch and on the impeller (Supplementary Fig. S12). As such, Mg impurities provide an ample number of nucleation sites in the solution, which can help maintain enhanced flowability and compressibility, providing significant advantages to LX processes. Furthermore, the transition to a single-crystalline-like morphology increased the filling ratio of the furnace during the cathode material calcination process, thereby enhancing process capacity (Supplementary Fig. S13).

Electrochemical performance verification

Conventionally, Mg^{2+} doping on layered oxide cathode was implemented via two different ways: (i) the use of separated Mg precursor (denoted as solid-state doping)[26] or (ii) the use of the Mg-containing TM precursor prepared through coprecipitation (denoted as coprecipitation doping)[27]. Here, we compared the electrochemical performances of these two conventional Mg^{2+} doping methods with a cathode from an impurity-adjusted lithium source (lithium-carbonate doping). All series of prepared Mg-containing Li_2CO_3 were used as lithium sources for synthesizing NCM622 cathodes, a widely used and representative layered-oxide-type cathode material. The powders were mixed with the transition metal precursor of $Ni_{0.6}Co_{0.2}Mn_{0.2}(OH)_2$ and calcinated (Supplementary Note S1; lithium-carbonate doping). The ICP-OES analysis (Supplementary Table S4) and XRD patterns (Supplementary Fig. S14) showed the exact Mg^{2+} concentration well doped in these cathodes from 0.5 to 3.6 mol% without detectable impurity phases, and negligible differences in particle morphology were observed (Supplementary Fig. S15).

Preliminarily electrochemical tests exhibited improved performances in cycle retention from the doping effect[28] with up to 1.5 mol% Mg-doping concentration (from a 1 wt% Mg-impurity-containing LCM precursor, denoted as LCD hereinafter). However, the further inclusion of impurity rather led to decreased performances (Supplementary Fig. S16). Given that the 1.5 mol% Mg-doped NCM622 showed the best electrochemical performance, the comparative doping study depending on synthesis routes was implemented at a doping concentration of 1.5 mol%. The synthesis routes of solid-state doping and coprecipitation doping were denoted as SSD and CPD, respectively (see detailed material characterization in Supplementary Note S3).

The comparison of electrochemical measurements revealed an interesting variation in the performance of Mg^{2+}-doped NCM622 cathodes depending on the doping methods (Fig. 3). Figure 3a, b respectively, illustrate the galvanostatic charge/discharge profiles of the first and second cycles at a current density of 16 mA g^{-1} in the voltage range of 3.0–4.3 V at 25 °C. The initial capacity was similar, with a slight decrease in the Mg^{2+}-doped cathodes, possibly attributable to the redox-inactive nature of Mg^{2+}. This consistency was observed at the higher temperature of 40 °C (Fig. 3c, d). All 1.5-mol% Mg^{2+}-doped cathodes exhibited improved cycling performances compared to the Bare cathode, with no significant initial capacity loss from multiple half-cell tests (Fig. 3e; error bars represent standard deviation). Notably, the LCD cathode achieved the highest capacity retention of 82.6%, while the SSD and CPD cathodes retained 80.9% and 78.8%, respectively, of their initial capacities after 200 cycles with a cutoff voltage of 4.3 V at a current density of 276.5 mA g^{-1} at 20 °C. To further accentuate the differences in capacity retention performance among the cathodes, high-temperature cycling tests were conducted at 40 °C accelerating cycling degradation. As shown in Fig. 3f, the LCD cathode retained over 60% of its capacity after 200 cycles, whereas the other cathodes reached 60% retention earlier. This cycling performance trend remained consistent with more pronounced differences at 40 °C. Additionally, the LCD cathode demonstrated superior rate capability compared with the other Mg-doping methods, as shown in rate capability tests conducted at various current densities with a cutoff voltage of 4.3 V at 25 °C (Fig. 3g). The full cells were constructed using commercial graphite as the anode to further verify the improved cycling performance of the LCD cathode (Fig. 3h). These full cells were cycled in the voltage range of 3.0–4.2 V (vs. graphite) at 40 °C, with a current density of 66.7 mA g^{-1}. The charge/discharge curves of the initial formation cycles of these full cells can be found in Supplementary Fig. 20. The comparable cathodes of LCD, SSD, CPD, and Bare retained 89.1%, 87.9%, 85.0%, and 80.8%, respectively, of their initial capacities after 200 cycles. The cathodes showed a consistent cycling performance trend in full cell composition, with LCD demonstrating the highest cycling stability. It should be highlighted that LCD, synthesized using a low-grade lithium source, outperformed other doping strategies without intensive impurity removal (further comparison studies for high-concentration Mg-doping are additionally conducted and described in Supplementary Fig. S19). Furthermore, the level of Mg impurity in the lithium precursor can easily be regulated by adjusting the refining process (see detailed explanation in Supplementary Note S4).

Anomalous doping effect of low-grade lithium sources

The electrochemical comparison suggests subtle doping state differences in cathode materials depending on the doping methods. Accordingly, fine crystal structural analysis was conducted using neutron diffraction (ND) to find the origin of such differences. It must be noted that ND is suitable for doping site studies because of its capability of detecting lightweight elements, such as Li and Mg, through nuclear interaction scattering[29]. Supplementary Fig. S22 compares the measured whole ND patterns of the bare and 1.5-mol% Mg^{2+}-doped NCM622s. We further simulated ND pattern changes according to two possible crystallographic positions of Mg^{2+}, exclusively in Li layers (*3a*) and TM layers (*3b*), and compared them with measured patterns (Fig. 4a, b). The pattern simulations (Fig. 4a and Supplementary Fig. S23A) show the peak height changes of the low-index planes of (003), (101), and (012) depending on the Mg^{2+} positions. With doping, the peak heights of the (003) and (101) planes became lower than those of the bare pattern regardless of positions, but the peak heights further weakened in the case of the *3a* site doping. A notable difference was found in the (012) peak, where the height decreased (−4.18%) as Mg^{2+} was located in the *3a* site but increased (+3.11%) in the *3b* site. Compared with the measured patterns, LCD followed more the tendency of the simulation result of Mg^{2+} in the *3a* site (Fig. 4b and Supplementary Fig. S23B), which showed the lowest peak heights at low-angle indices at (003), (101), and (012). Meanwhile, the SSD and CPD patterns matched more with the Mg^{2+} in the TM site simulation results.

The further combination of XRD (Supplementary Fig. S24) and ND refinement analysis gave more solid evidence for the selective Mg-doping phenomenon. For increased accuracy, powder XRD refinement first provided the general information of the samples, such as the lattice parameters and atomic positions, and the following ND analysis strictly confirmed the atomic occupancy and thermal coefficient on the Li slab separating Li^+, Mg^{2+}, and Ni^{2+} signals using their distinguishable atomic scattering in ND. Then, this XRD–ND iterative refinement was implemented until all refinement indicators were in reliable ranges. All Rietveld refinement results of the fitting data and detailed refinement parameters are in Supplementary Figs. S25 and S26 and Tables S8–S12. The most noticeable difference was that LCD

https://doi.org/10.1038/s41467-024-44812-3

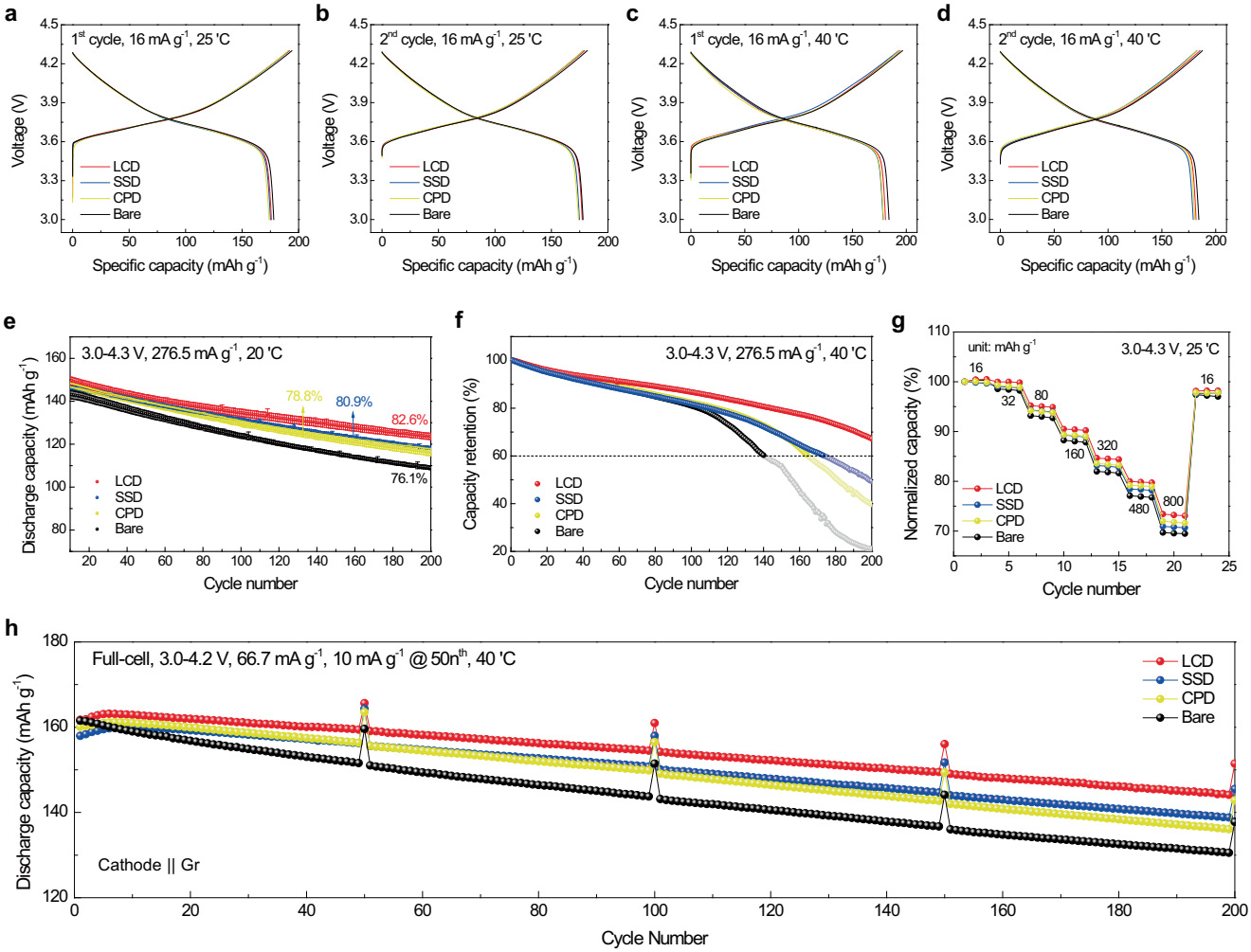

Fig. 3 | Comparison of electrochemical performances depending on doping strategies. a–d Charge/discharge curves of LCD, SSD, CPD, and Bare in the voltage range of 3.0–4.3 V at a current density of 16 mA g^{-1}; first cycle (**a**) and second cycle (**b**) at 25 °C, first cycle (**c**) and second cycle (**d**) at 40 °C. **e**, **f** Capacity fading of cathodes at a current density of 276.5 mA g^{-1} with a cutoff voltage of 4.3 V at 20 °C (**e**) and 40 °C (**f**). **g** Rate capability of cathodes at various current rates of with a cutoff voltage of 4.3 V at 25 °C. **h** Capacity fading of cathodes at a current density of 66.7 mA g^{-1} with a cutoff voltage of 4.2 V (vs. graphite) at 40 °C in the full cell. The capacities were checked every 50th cycle at a current density of 10 mA g^{-1}, and a constant voltage of 3.0 V was applied at the end of discharge (10 mA g^{-1} current cutoff in the constant voltage).

showed the highest occupancy ratio of Mg^{2+} in the Li site (Mg_{Li}%), as summarized in Fig. 4c. More than 80% of doped Mg^{2+} ions, corresponding to 1.2 mol%, occupied the Li site in LCD, whereas around 65% of Mg^{2+} ions were found in the Li slab for both SSD and CPD. Furthermore, X-ray absorption results (Supplementary Fig. S27) supported the doping site selectivity of Mg depending on the synthetic route (more detailed discussions about other refinement parameters are in Supplementary Fig. S28). The Mg^{2+} ions in Li layers resulted in the so-called pillaring effect between TM layers, which enhanced cyclability and rate capability (Fig. 3) by hindering the lattice collapse and increasing the *c*-axis spaces of Li diffusion layers, respectively[27,28,30,31]. Moreover, a previous study that selectively doped Mg^{2+} ions into Li sites minimized the replacement of redox-active TMs with Mg^{2+} and thus minimized the initial capacity loss[32]. These arguments will explain the superior electrochemical performances of LCD compared with those of other doping methods (additional structural studies of high-concentration Mg^{2+}-doped cathode materials can be found in Supplementary Note S5).

When the final doping state solely follows thermodynamics, the concentrations at certain doping sites would be identical, regardless of the synthesis methods. However, differences in precursor mixtures or material conditions can lead to variations in the final product results due to distinct kinetic environments. For instance, recent research has reported that the kinetics of lithium transition metal layered oxide growth noticeably varied depending on the distribution of the Li source around the transition metal oxide host structure[33]. Such findings introduce a new perspective on how the thermal diffusion of doping or guest ions into the host structure can significantly influence the final phase. It is well known that the TM hydroxide precursor ($Ni_{0.6}Co_{0.2}Mn_{0.2}(OH)_2$) turns into a rocksalt structure of (Ni, Co, Mn)–O after the water inside the precursor evaporates. Thereafter, Li ions thermally diffuse into the rocksalt host structure above the melting point of the Li source (e.g., 723 °C for Li_2CO_3), which oxidizes TM and constructs an ordered layered structure of cathode material (Li + $Ni_{0.6}Co_{0.2}Mn_{0.2}O_2$)[34,35]. These previous insights naturally lead to the consideration of the thermal diffusion behaviour of doping and guest ions, with expected kinetic differences for Mg^{2+} and Li^+ due to the melting points of $MgCO_3$ (350 °C) and Li_2CO_3 (723 °C), respectively. The thermal diffusion behaviour of SSD and LCD was confirmed through chemical composition analysis. The precursor mixture of SSD and LCD was calcined at 400 °C for 5 h, which was above the melting point of $MgCO_3$ but below the melting point of Li_2CO_3, in order to analyse the intermediate phase. The backscattered electron image (Supplementary Fig. S32) shows sufficient mechanical mixing of Li and

https://doi.org/10.1038/s41467-024-44812-3

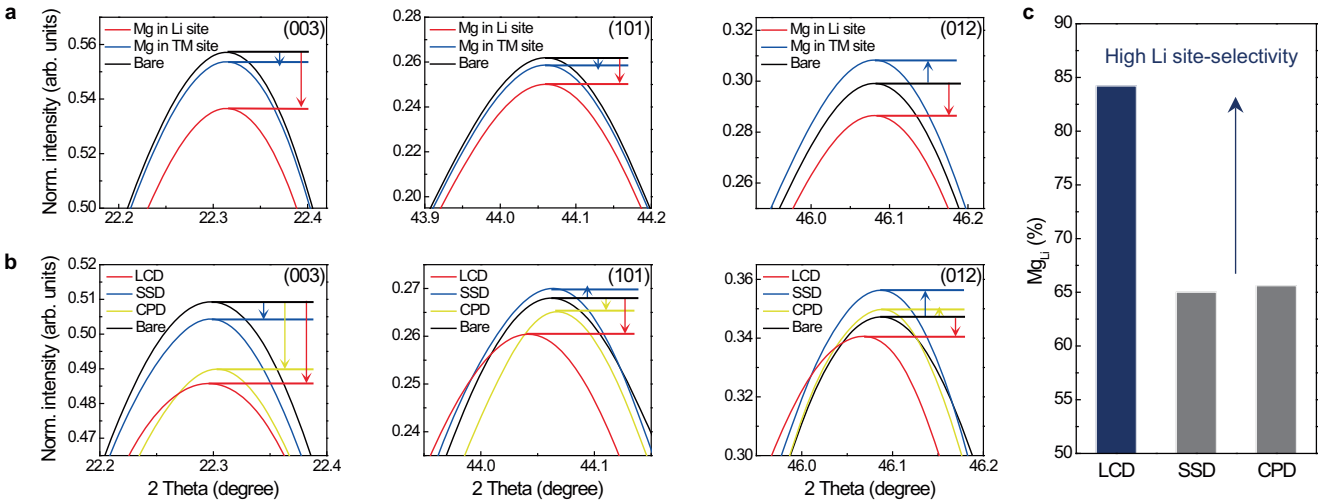

Fig. 4 | Fine structural analysis defining the crystallographic position of Mg. a, **b** Comparisons of simulated (**a**) and observed (**b**) ND patterns. **c** The ratio of Mg in the Li slab against the total amounts of Mg in the cathode materials was analysed through ND Rietveld refinement.

TM precursors. SEM-EDS analysis further confirmed that Mg was homogeneously distributed in all SSD particles (Supplementary Figs. S33 and S34), while Mg and TMs were distributed in different spatial regions in the LCD mixture under identical conditions (Supplementary Figs. S33 and S35). Additionally, the altered XRD pattern of the intermediated rocksalt phase provided clear evidence of the presence of Mg in the rocksalt phase of SSD (Supplementary Fig. S36). This observation directly demonstrates that the synthesis results could vary depending on which precursor contains the dopant.

On the basis of the above findings, we propose doping models depending on the sources of Mg (Fig. 5). Let us consider the doping processes of SSD and CPD incorporating Mg^{2+} ions from separated or transition metal precursors, respectively. The formation of the (Ni, Co, Mn)–O rocksalt structure initially occurred at a certain temperature in the SSD mixture (Fig. 5a). Then, $MgCO_3$ is liquified above its melting point of 350 °C, and Mg^{2+} ions diffused into (Ni, Co, Mn)–O to form the (Ni, Co, Mn, Mg)–O rocksalt phase. Likewise, the TM hydroxide precursor of CPD (Fig. 5b) formed the (Ni, Co, Mn, Mg)–O rocksalt structure as the water evaporated. The (Ni, Co, Mn, Mg)–O rocksalt phases in SSD and CPD constructed the final ordered layered phases above the melting point of the Li source as Li ions were thermally supplied and Mg^{2+} ions were redistributed under the constructed layered structure.

Meanwhile, the LCD (Fig. 5c) case was unable to uptake Mg^{2+} ions into its rocksalt (Ni, Co, Mn)–O phase until the Mg/Li source was liquified. The TM hydroxide precursor formed (Ni, Co, Mn)–O at an elevated temperature, but the solid solution of dilute $MgCO_3$ in Li_2CO_3 (Fig. 2a–d) more followed the melting behaviour of the major phase when considering the typical phase diagram of binary solid-solution systems (Supplementary Fig. S37). As the temperature increased near ~723 °C, Li^+ and Mg^{2+} ions were then able to co-diffuse into the (Ni, Co, Mn)–O rocksalt structure simultaneously, forming the final layered phase where most of the Mg^{2+} ions were positioned at the Li site.

According to a previous work, Mg^{2+} ions thermodynamically prefer Li sites at concentrations below 2 mol%[30]. Nevertheless, the actual doping result varied sensitively depending on synthetic conditions. Thus, we must consider the melting point of the dopant precursor and the degree of distribution of the doping element. The site selectivity of the dopant was kinetically controlled by adjusting the incorporation timing of the doping element. Because of the solid-solution property of Mg^{2+}-containing Li_2CO_3, Mg^{2+} could be trapped in the Li_2CO_3 until the layered phase evolution begins, whereas solid-state doping and coprecipitation doping processed first the formatting of Mg^{2+}-containing rocksalt structure.

Economic and environmental impacts

The inclusion of Mg impurity during lithium extraction improved the lithium extraction rate, powder flowability, and compressibility through controlled particle morphology. Moreover, the fine structural analysis demonstrated that Mg impurity at a level of about 1% in the lithium source can boost the electrochemical performance by maximizing the pillaring effect and increasing the layer space. This comprehensive re-evaluation of impurity grade provides us a chance to address the cost and environmental issues of the production of battery materials. In this regard, we calculated an annual plant production capacity of 25,000 tons of Li_2CO_3 per year. On this basis, economic and environmental evaluations were conducted for the introduction of 1% Mg impurity low-grade Li_2CO_3 (Table 1; more detailed calculation tables are in Supplementary Tables S13–S21). It is worth noting that the pilot plant (P/P) test conducted here utilized actual brine pumped from 'Hombre Muerto' Salt Lake in the Argentine highlands and spodumene mined from Western Australia of which results served as the basis for the evaluations.

The economic evaluation was performed in two respects: (i) capital expenditure (CAPEX); productivity and the scale of facilities and (ii) operating expenditure (OPEX). The CAPEX was mainly attributed to solid–liquid filtration (S/L filtration) equipment cost for eliminating $Mg(OH)_2$ by-products (Supplementary Tables S15 and S16). As shown in our P/P test (Supplementary Table S17), the Mg refinery generated 15.84 ton (LC ton)$^{-1}$ of $Mg(OH)_2$ cake, whereas the Ca refinery produced only 0.98 ton (LC ton)$^{-1}$ of $CaCO_3$. Moreover, as the D10 of the $Mg(OH)_2$ cake was generally obtained around 3 μm or below (Supplementary Fig. S38), this small-sized $Mg(OH)_2$ precipitation resulted in a cost issue in the S/L filtration process. The small amount of impurity was hard to remove using normal filter press equipment; thus, additional ion exchange filters (IX) or advanced disc filter processes are required as the next purifying step to achieve battery-grade purity. Here, however, the advanced filtration step is now able to be skipped with improved Mg impurity tolerance (~1%), which led to a considerable reduction in equipment cost for brine and hard rock processes of 12 and 9.5 million USD, corresponding to 19.4% and 7.3% reductions from total equipment cost, respectively (Table 1, Supplementary Tables S15 and S16).

With higher Mg impurity tolerance, the total OPEX costs were significantly reduced as well in both brine and hard rock processes by